

Tutorial 5

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DATE			
PAGE No			

① compute the following:

① A 5×5 input feature map is convolved with a 3×3 filter using a stride of 1 and no {valid} padding. what is the output shape?

Given,

Input size = 5×5

filter size = 3×3

stride (S) = 1

padding (P) = 0

Formula:

$$\text{Output size} = \frac{N - F + 2P}{S} + 1$$

$$= \frac{5 - 3 + 2(0)}{1} + 1$$

$$= \frac{2}{1} + 1$$

$$= 2 + 1 = 3$$

So, output height = 3 and output width = 3

final answer: 3×3

- ② A 6×6 input feature map is processed using a 3×3 convolutional filter with a stride of 1 and "same" padding. What is the output shape?

Given:

Input size = 6×6

filter size = 3×3

stride $S = 1$

same padding

formulas

$$W_{out} = \frac{W_{in} - F + 2P}{S} + 1$$

$$H_{out} = \frac{H_{in} - F + 2P}{S} + 1$$

With same padding and stride = 1, the output size remains the same as the input size. The sheet states that same padding keeps the output the same size as input.

so, $W_{out} = 6$

$H_{out} = 6$

~~the output shape is 6×6~~

- (3) A 7×7 input undergoes a 5×5 convolution with a stride of 2 and padding of $P=1$.

given,

$$\text{Input size} = 7 \times 7$$

$$\text{Filter size} = 5 \times 5$$

$$\text{Stride } S = 2$$

$$\text{Padding } P = 1$$

Formula,

$$W_{out} = \frac{W_{in} - F + 2P}{S} + 1$$

$$H_{out} = \frac{H_{in} - F + 2P}{S} + 1$$

substitute the values!

$$W_{out} = \frac{7 - 5 + 2(1)}{2} + 1$$

$$= \frac{7 - 5 + 2}{2} + 1$$

$$= \frac{4}{2} + 1$$

$$= 2 + 1 = 3 //$$

similarly, $H_{out} = \frac{7 - 5 + 2(1)}{2} + 1 = 3$

Final Answers: $3 \times 3 //$

Q. An 8×8 input is processed with a 5×5 filter using a stride of 2 and no padding.

Given,

$$\text{Input size} = 8 \times 8$$

$$\text{filter size} = 5 \times 5$$

$$\text{stride } s = 2$$

$$\text{padding } p = 0, \text{ no padding,}$$

Formula,

$$W_{out} = \frac{W_{in} - F + 2P}{s} + 1$$

$$H_{out} = \frac{H_{in} - F + 2P}{s} + 1$$

Substitute the values,

$$W_{out} = \frac{8 - 5 + 2(0)}{2} + 1$$

$$= \frac{8 - 5}{2} + 1$$

$$= \frac{3}{2} + 1$$

$$= 1 + 1 = 2 //$$

Similarly,

$$H_{out} = \frac{8 - 5 + 2(0)}{2} + 1 = 2 //$$

Final answer: 2×2 //

Denan Wanem

DATE			
PAGE No.			

- ⑤ A 10×5 input is convolved using a 3×3 filter with a stride of 1 and "same" padding.

Given,

$$\text{Input size} = 10 \times 5$$

$$\text{filter size} = 3 \times 3$$

$$\text{Stride } S = 1$$

$$\text{padding} = \text{same}$$

Formula,

$$W_{out} = \frac{W_{in} - F + 2P}{S} + 1$$

$$H_{out} = \frac{H_{in} - F + 2P}{S} + 1$$

With same padding and stride = 1, the output size remains the same as the input size. The checker states that same padding keeps the output the size as output -

So,

$$W_{out} = 10$$

$$H_{out} = 5$$

final Answer, 10×5

(3) Max pooling
Given convolution output,

$$C = \begin{bmatrix} 225 & 258 & 250 & 209 \\ 458 & 566 & 552 & 472 \\ 708 & 981 & 887 & 802 \\ 1000 & 1488 & 1320 & 1224 \end{bmatrix}$$

Given,

Window size = 2×2

stride = 2

Step 1: output dimension

$$W_{out} = \frac{W_{in} - F}{S} + 1$$

$$H_{out} = \frac{H_{in} - F}{S} + 1$$

Substituting

$$W_{out} = \frac{4-2}{2} + 1 = 2$$

$$H_{out} = \frac{4-2}{2} + 1 = 2$$

so, output size is 2×2

After max pooling,
first block:

$$\begin{bmatrix} 225 & 258 \\ 458 & 566 \end{bmatrix} \Rightarrow 566$$

Darin Wanem

DATE

PAGE No.

$$\text{Second block} : \begin{bmatrix} 250 & 209 \\ 552 & 492 \end{bmatrix} \rightarrow 552$$

$$\text{Third block} = \begin{bmatrix} 708 & 981 \\ 1000 & 1488 \end{bmatrix} = 1488$$

$$\text{Fourth block} = \begin{bmatrix} 287 & 202 \\ 1320 & 1214 \end{bmatrix} = 1320$$

$$\text{Final Answer} = M = \begin{bmatrix} 566 & 552 \\ 1488 & 1320 \end{bmatrix}$$

q. Average Pooling

Given,

$$X = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$$

Given,

filter size $> 2 \times 2$

stride $= 2$

step 1! output dimension,

$$W_{out} = \frac{4-2}{2} + 1 = 2$$

$$H_{out} = \frac{4-2}{2} + 1 = 2$$

So, Output size is! 2×2

Dehan Wanem

DATE			
PAGE NO			

Step 2) Apply average pooling:

First block: $\begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix} = \frac{1+2+5+6}{4} = \frac{14}{4} = 3.5$

second block: $\begin{bmatrix} 3 & 4 \\ 7 & 8 \end{bmatrix} \Rightarrow \frac{3+4+7+8}{4} = \frac{22}{4} = 5.5$

Third block: $\begin{bmatrix} 9 & 10 \\ 13 & 14 \end{bmatrix} = \frac{9+10+13+14}{4} = \frac{46}{4} = 11.5$

Fourth block: $\begin{bmatrix} 11 & 12 \\ 15 & 16 \end{bmatrix} = \frac{11+12+15+16}{4} = \frac{54}{4} = 13.5$

Final Answer:

Mavg = $\begin{bmatrix} 3.5 & 5.5 \\ 11.5 & 13.5 \end{bmatrix}$ //